

Coding & Solution

Date	9 July 2026
Team ID	
Project Name	Credit Card Approval Prediction System
Maximum Marks	5 Marks

Coding & Solution

This document evaluates the implementation of the Credit Card Approval Prediction System. The project is developed using Python, Flask, and Machine Learning to automate the credit card approval process. It demonstrates clean coding practices, modular design, reusable components, proper documentation, and alignment with the functional and non-functional requirements defined during the planning phase.

Instructions

1. The project code follows a modular architecture with separate files for model training, prediction, and web application development.
2. The solution addresses the core problem statement by predicting whether a credit card application will be approved or rejected based on applicant information.
3. The project contains all required source files, configuration files, datasets, trained models, HTML templates, CSS files, and setup instructions.
4. The complete source code is maintained in a GitHub repository for version control and future enhancements.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/yellamandakiran/CreditCardApprovalPrediction URL- https://creditcardapprovalprediction-7xg5.onrender.com
Programming Language(s)	Python, HTML5, CSS3, JavaScript
Framework(s) Used	Flask, Bootstrap, Scikit-learn, XGBoost
Key Features Implemented	• Dataset preprocessing and cleaning. • Missing value handling. • Duplicate removal. • Feature engineering. • Label encoding. • Logistic Regression model. • Decision Tree model. • Random Forest model. • XGBoost model. • Model evaluation. • Best model selection. • Flask web application. • User-friendly prediction interface. • Real-time approval prediction. • Model serialization using Joblib. • Ready for IBM Watson Machine Learning deployment.
Pending / Incomplete Features	• User authentication. • Prediction history dashboard. • Email notifications. • Database integration. • Explainable AI feature. • Admin analytics dashboard.
Setup / Run Instructions	1. Install Python 3.10 or above. 2. Install required packages using <code>pip install -r requirements.txt</code>. 3. Place <code>application_record.csv</code> and <code>credit_record.csv</code> inside the dataset folder. 4. Run <code>python train_model.py</code> to train and save the model. 5. Run <code>python app.py</code>. 6. Open <code>http://127.0.0.1:5000</code> in your browser. 7. Enter applicant details and click Predict.

Code Quality Checklist

S.No	Criteria	Status (Yes/No)
1	Code is modular and organized into functions/classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments/documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

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